A STUDY OF RANDOM MATRICES

Sample Covariance Matrices and Spectral Clustering

AN INDEPENDENT STUDY REPORT

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Abstract

[TODO] *Write the abstract once the body is finalized. Suggested skeleton: (1) Open with a one-sentence framing of random matrix theory and why sample covariance matrices matter in high-dimensional statistics. (2) State the scope: we focus on Gaussian sample covariance matrices in Johnstone's spiked model. (3) Outline the journey: foundational probability and linear algebra, sample covariance matrices and the Wishart distribution, low-dimensional fluctuation behavior, the high-dimensional Marchenko-Pastur and Tracy-Widom laws, the spiked covariance model and the BBP phase transition, eigenvector alignment beyond the threshold, and a sketch of the connection to spectral clustering. (4) Note the role of Monte Carlo simulations in illustrating each phase of the argument. (5) Close with the takeaway: the contrast between strong and weak spikes produces a clean phase transition in both eigenvalue location and eigenvector informativeness, with implications for spectral clustering. Avoid em dashes; use clean academic prose.*

Acknowledgements

[TODO] *Acknowledgements paragraph. Suggested mentions: Professor Gustavo Didier (advisor) for the weekly meetings, the reading recommendations (Yao, Zheng, and Bai 2015; Féral and Péché 2009), and the patient guidance through the spiked covariance literature. The second reader (TBD) for serving on the committee. Optional: family, friends, and the Tulane Department of Mathematics.*

Contents

1 Introduction

Why random matrices?

Modern data sets are large and high-dimensional. A single-cell genomics experiment may measure tens of thousands of gene expression levels in a few hundred cells. A finance application may track hundreds of asset returns over a few years of daily data. A computer-vision system may compare millions of image patches in a feature space of several hundred dimensions. In each case the number of variables p is comparable to, or larger than, the number of observations n. The most basic question one can ask of such data is how its variance is distributed across orthogonal directions, and which directions explain the most variation. Answering this question is the job of *principal components analysis*, and the algorithm reduces to computing the eigenvalues and eigenvectors of the *sample covariance matrix*

$$\widehat{\Sigma}_n \;=\; \frac{1}{n}\sum_{i=1}^{n} \boldsymbol{X}_i \boldsymbol{X}_i^\top, \tag{1}$$

where the $\boldsymbol{X}_i \in \mathbb{R}^p$ are the n observed data points, assumed centered.

Classical multivariate statistics tells us what to expect when p is small and n is large. In this low-dimensional regime, $\widehat{\Sigma}_n$ is an accurate estimate of the population covariance Σ, its eigenvalues converge to the population eigenvalues, and its eigenvectors converge to the population eigenvectors. Methods that depend on this convergence, from PCA to factor analysis to the *spectral clustering* algorithms that we will reach at the end of this report, then stand on firm ground.

The classical picture breaks down when p and n are comparable. The sample covariance matrix is no longer a reliable estimate of Σ: its eigenvalues spread into a non-trivial band of random values, and its eigenvectors may carry little or no information about the population directions. Figure 1 makes the contrast concrete. The same Gaussian sampling model produces eigenvalues that concentrate at the population value 1 when $p = 10$ and $n = 1000$, and eigenvalues that spread across a wide interval, with a well-defined limiting density, when $p = 500$ and $n = 1000$.

The mathematical framework that explains and predicts this phenomenon is *random matrix theory*, originally developed for problems in nuclear physics in the 1950s and now a central tool in high-dimensional statistics, signal processing, and machine learning (Yao et al., 2015; Vershynin, 2018; Couillet and Liao, 2022).

What this report does, and how

This report develops, from first principles, the spectral analysis of sample covariance matrices in both regimes, and uses what we learn to understand when methods like spectral clustering can recover hidden structure in high-dimensional data. The development is built bottom-up: every claim is either proved in the text, sketched with a pointer to a full proof, or cited with the cleanest reference we could find. Every theoretical claim is then illustrated with a Monte Carlo simulation in Python; the notebooks live in the `notebooks/` directory of

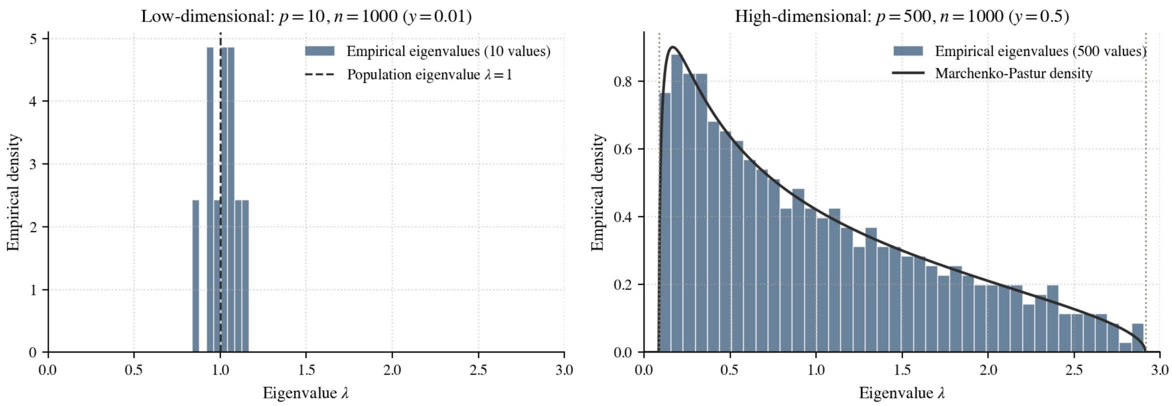

Figure 1: Eigenvalues of the sample covariance matrix $\widehat{\Sigma}_n$ when $\Sigma = I_p$ and the data are Gaussian, for two aspect ratios. *Left:* when $p = 10$ and $n = 1000$ (so $p/n = 0.01$), the ten sample eigenvalues concentrate tightly around the population value $\lambda = 1$, confirming the classical low-dimensional intuition. *Right:* when $p = 500$ and $n = 1000$ (so $p/n = 0.5$), the five hundred sample eigenvalues spread across the interval $[(1 - \sqrt{y})^2, (1 + \sqrt{y})^2] \approx [0.086, 2.914]$. The empirical density agrees with the Marchenko-Pastur prediction (solid curve). The phenomenon that motivates this report is already visible: in high dimensions, sample eigenvalues are biased, and the spread is a real feature of the geometry, not measurement noise.

the accompanying repository, and the figures they produce are reproduced alongside the corresponding statements.

The starting point is the random variable, the basic building block of probability theory. From there we build to moments and moment generating functions, characteristic functions, modes of convergence, the central limit theorem, random vectors with multivariate normal distributions, and the spectral theorem for symmetric matrices. Each of these objects feeds directly into the analysis of $\widehat{\Sigma}_n$ that follows.

The story arc

The mathematical journey unfolds in three movements.

1. *The low-dimensional regime* (p fixed, $n \to \infty$). The law of large numbers gives entrywise convergence of $\widehat{\Sigma}_n$ to Σ, and continuity of the characteristic polynomial then forces the sample eigenvalues to converge to the population eigenvalues. The size and shape of the fluctuations around these limits depend on whether the population eigenvalues are simple or repeated. For $\Sigma = I_p$ the matrix $\sqrt{n}(\widehat{\Sigma}_n - I_p)$ converges in distribution to a *Gaussian Orthogonal Ensemble* matrix, whose top eigenvalue has a non-Gaussian limit shaped by *eigenvalue repulsion*. For Σ with distinct eigenvalues, the delta method instead produces ordinary Gaussian fluctuations of each sample eigenvalue around its population counterpart.

2. *The high-dimensional regime* ($p, n \to \infty$ with $p/n \to y \in (0, \infty)$). Even for the null covariance $\Sigma = I_p$, the sample eigenvalues no longer concentrate at 1; they fill out the *Marchenko-Pastur law* on the interval $[(1 - \sqrt{y})^2, (1 + \sqrt{y})^2]$ (Marchenko and Pastur,

1967), exactly as in the right panel of Figure 1. The largest eigenvalue converges almost surely to the upper edge $(1+\sqrt{y})^2$, but its fluctuations around that edge are not Gaussian. They live on the scale $p^{-2/3}$ and follow the *Tracy-Widom law* (Tracy and Widom, 1996; Johnstone, 2001), a distinct universality class first discovered in random matrix theory and now found in problems ranging from longest increasing subsequences to growth models in mathematical physics.

3. *Structure and a phase transition.* When the population covariance carries low-rank structure, a small number of distinguished eigenvalues riding above a flat baseline, a striking new phenomenon appears. If a signal eigenvalue exceeds a threshold determined by the aspect ratio y, an isolated outlier eigenvalue appears in the sample spectrum and the corresponding sample eigenvector aligns nontrivially with the population spike direction. If the signal is below the threshold, the spike is absorbed into the Marchenko-Pastur bulk and its eigenvector becomes asymptotically orthogonal to the population direction. This phase transition is the bridge to spectral clustering: the same picture explains how clustering algorithms recover hidden class structure from data, but only when the underlying signal-to-noise ratio crosses an analogous threshold.

Where this is applicable

The techniques developed here are not abstract curiosities. The Marchenko-Pastur law sets a baseline for detecting structure in any high-dimensional covariance estimate, with applications in array-signal processing, wireless communications, and quantitative finance (Couillet and Liao, 2022). The Tracy-Widom edge governs the null distribution of the largest principal component, which is the basis for principled testing of how many genuine signal directions are present in a data set (Johnstone, 2001). The spike phase transition determines when a low-rank signal in noise is recoverable by PCA, with direct implications for denoising and dimensionality reduction. Spectral clustering, finally, is one of the most widely used unsupervised methods for problems where the natural geometry is not Euclidean, including image segmentation, community detection in networks, and clustering of single-cell data (von Luxburg, 2007; Ng et al., 2002). The phase transition we develop in the spiked model is the cleanest mathematical lens we have for understanding when these methods succeed and when they fail.

Roadmap

Section 2 reviews probability foundations: random variables, moments and moment generating functions, characteristic functions, modes of convergence, the law of large numbers and central limit theorem, and random vectors. Section 3 develops the sample covariance matrix in both regimes, from the spectral theorem and the Wishart distribution in low dimensions through the Marchenko-Pastur and Tracy-Widom laws in high dimensions. Section 4 introduces the spiked covariance model, proves the eigenvalue phase transition, derives the eigenvector overlap formula, and sketches the connection to spectral clustering. Section 5 concludes with a summary and open questions, and three appendices collect auxiliary probabilistic results, simulation methodology, and selected Python code listings.

2 Probability Foundations

The high-dimensional phenomena studied in later sections all rest on a small set of foundational ideas from classical probability: distributions of random variables, ways to summarize and compare them, modes of convergence, the law of large numbers, and the central limit theorem. This section develops these tools in the depth needed to make later proofs go through cleanly. The exposition starts with the random variable, the basic object of probability theory, and builds toward the multivariate normal distribution that will drive every Gaussian sample-covariance computation in Section 3.

2.1 Random Variables and Distributions

Probability begins with a *probability space* $(\Omega, \mathcal{F}, \mathbb{P})$, where Ω is the set of possible outcomes, $\mathcal{F}$ is a collection of events (a σ-algebra of subsets of Ω), and $\mathbb{P}$ is a probability measure assigning each event a number in $[0, 1]$. We work with this triple in the background; the foreground objects are real-valued functions of the outcome.

Definition 2.1 (Random variable)**.** A *random variable* is a function $X : \Omega \to \mathbb{R}$. Its distribution is captured by the *cumulative distribution function* (CDF)

$$F_X(x) \;=\; \mathbb{P}(X \leq x), \qquad x \in \mathbb{R}.$$

A CDF is determined by three properties.

Proposition 2.1 (Characterization of CDFs)**.** The CDF of any random variable satisfies:

(F1) $\lim\limits_{x \to -\infty} F_X(x) = 0$ and $\lim\limits_{x \to +\infty} F_X(x) = 1$;

(F2) F_X is right-continuous: $\lim_{x \to a^+} F_X(x) = F_X(a)$;

(F3) F_X is non-decreasing: $a \leq b \Rightarrow F_X(a) \leq F_X(b)$.

Conversely, any $F : \mathbb{R} \to [0, 1]$ satisfying (F1)–(F3) is the CDF of some random variable (Casella and Berger, 2002, §1.5).

Random variables come in two main flavors. *Discrete* random variables take values in a countable set $\{x_1, x_2, \ldots\}$, and their distribution is described by the *probability mass function* (PMF) $p_X(x_i) = \mathbb{P}(X = x_i)$. *Absolutely continuous* random variables admit a nonnegative *probability density function* (PDF) f_X with

$$F_X(x) \;=\; \int_{-\infty}^{x} f_X(t)\, dt.$$

Two random variables X, Y are *independent* if $\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\, \mathbb{P}(Y \in B)$ for all Borel sets $A, B \subseteq \mathbb{R}$.

Example 2.2 (Distributions used throughout)**.** The following appear repeatedly in later sections:

- *Bernoulli*(p): $X \in \{0,1\}$ with $\mathbb{P}(X=1)=p$.
- *Poisson*(λ): $\mathbb{P}(X=k)=e^{-\lambda}\lambda^k/k!$ for $k=0,1,2,\ldots$.
- *Normal* $\mathcal{N}(\mu,\sigma^2)$: $f_X(x)=(\sqrt{2\pi}\,\sigma)^{-1}\exp\bigl(-(x-\mu)^2/(2\sigma^2)\bigr)$.
- *Exponential*(λ): $f_X(x)=\lambda e^{-\lambda x}$ for $x \geq 0$.
- *Cauchy*: $f_X(x)=1/\bigl(\pi(1+x^2)\bigr)$.

The Cauchy distribution will be a recurring foil: its heavy tails make some standard tools break, motivating sharper tools that do not.

2.2 Moments and Moment Generating Functions

Moments quantify the shape of a distribution.

Definition 2.3 (Moments and central moments)**.** For a random variable X with finite $\mathbb{E}|X|^j$, the j^{th} *moment* is $\mathbb{E}[X^j]$ and the j^{th} *central moment* is $\mathbb{E}[(X-\mu)^j]$ where $\mu=\mathbb{E}[X]$. The second central moment is the *variance*, $\operatorname{Var}(X)=\mathbb{E}[(X-\mu)^2]$.

A generating function for the moments compresses all of this information into a single function.

Definition 2.4 (Moment generating function)**.** The *moment generating function* (MGF) of X is

$$M_X(t) \;=\; \mathbb{E}[e^{tX}], \qquad t \in \mathbb{R},$$

provided the expectation is finite on an open neighborhood $(-h,h)$ of zero.

The name is justified by the following.

Theorem 2.2 (MGFs generate moments)**.** If $M_X(t)$ is finite for $t \in (-h,h)$, then M_X is infinitely differentiable at 0 and

$$M_X^{(n)}(0) \;=\; \mathbb{E}[X^n], \qquad n=1,2,\ldots$$

Proof. Consider the continuous case; the discrete case is analogous, with sums replacing integrals. Finiteness of M_X on $(-h,h)$ justifies differentiation under the integral sign (Casella and Berger, 2002, Thm. 2.4.1), so

$$\frac{d}{dt}M_X(t) \;=\; \int_{-\infty}^{\infty}\frac{d}{dt}\bigl(e^{tx}\bigr)f_X(x)\,dx \;=\; \int_{-\infty}^{\infty} xe^{tx}f_X(x)\,dx \;=\; \mathbb{E}[Xe^{tX}].$$

Evaluating at $t=0$ gives $M_X'(0)=\mathbb{E}[X]$. Iterating the same argument n times yields $M_X^{(n)}(t)=\mathbb{E}[X^n e^{tX}]$, and hence $M_X^{(n)}(0)=\mathbb{E}[X^n]$. ∎

A second property is the source of most computational power: the MGF of a sum of independent random variables is the product of their MGFs. Indeed, if X and Y are independent,

$$M_{X+Y}(t) \;=\; \mathbb{E}[e^{t(X+Y)}] \;=\; \mathbb{E}[e^{tX}]\,\mathbb{E}[e^{tY}] \;=\; M_X(t)\,M_Y(t).$$

We compute MGFs of the canonical distributions to make this concrete.

Example 2.5 (MGFs of standard distributions)**.** *Bernoulli*(p):

$$M_X(t) \;=\; (1-p)e^0 + pe^t \;=\; 1-p+pe^t.$$

Poisson(λ): substituting the PMF and recognizing the Taylor series of exp,

$$M_X(t) \;=\; \sum_{k=0}^{\infty} e^{tk}\, e^{-\lambda}\,\frac{\lambda^k}{k!} \;=\; e^{-\lambda}\sum_{k=0}^{\infty}\frac{(\lambda e^t)^k}{k!} \;=\; \exp\bigl(\lambda(e^t-1)\bigr).$$

Normal $\mathcal{N}(\mu,\sigma^2)$: substituting the PDF and completing the square in the exponent $tx - (x-\mu)^2/(2\sigma^2)$ yields a Gaussian kernel that integrates to $\sqrt{2\pi}\,\sigma$, giving

$$M_X(t) \;=\; \exp\Bigl(\mu t + \tfrac{1}{2}\sigma^2 t^2\Bigr), \qquad t\in\mathbb{R}.$$

These three examples illustrate the technique. The multiplicative property turns difficult convolutions into easy multiplications: if $X_1,\dots,X_n$ are iid Bernoulli(p), their sum has MGF $(1-p+pe^t)^n$, which is the MGF of Binomial(n,p); independent Poissons of rates λ_1,λ_2 sum to a Poisson of rate $\lambda_1+\lambda_2$; independent normals sum to a normal with summed mean and summed variance.

Beyond computing moments, the MGF determines the distribution uniquely.

Theorem 2.3 (MGF uniqueness)**.** If $M_X(t) = M_Y(t)$ for all t in some neighborhood $(-h,h)$ of zero, then X and Y have the same distribution.

The proof rests on the uniqueness of two-sided Laplace transforms; see Casella and Berger (2002, Thm. 2.3.11). The converse direction, that equal distributions have equal MGFs, is immediate from $\mathbb{E}[g(X)] = \mathbb{E}[g(Y)]$ applied with $g(x) = e^{tx}$.

There is a catch. The MGF requires e^{tX} to have a finite expectation for t near zero, which can fail if X has heavy tails.

Example 2.6 (Cauchy: no MGF)**.** For the standard Cauchy, $\int_{-\infty}^{\infty} e^{tx}\,/(\pi(1+x^2))\,dx$ diverges for every $t\neq 0$. The MGF therefore does not exist on any open neighborhood of zero. This is not a pathology; the Cauchy is a perfectly well-defined distribution, central in probability and physics.

The remedy is the characteristic function.

2.3 Characteristic Functions

Replacing the real argument t of the MGF by the imaginary argument it yields an object that always exists.

Definition 2.7 (Characteristic function)**.** The *characteristic function* (CF) of a random variable X is

$$\varphi_X(t) \;=\; \mathbb{E}[e^{itX}] \;=\; \mathbb{E}[\cos(tX)] + i\,\mathbb{E}[\sin(tX)], \qquad t\in\mathbb{R}.$$

Because $|e^{itx}| = 1$ for every real x, the integrand is bounded, and so

$$|\varphi_X(t)| \;\le\; \mathbb{E}[|e^{itX}|] \;=\; 1, \qquad \varphi_X(0) = 1.$$

The CF is therefore well defined for every distribution. When X has a density, φ_X is exactly the Fourier transform of the density:

$$\varphi_X(t) \;=\; \int_{-\infty}^{\infty} e^{itx}\, f_X(x)\, dx.$$

When the MGF exists, formal substitution gives $\varphi_X(t) = M_X(it)$.

Example 2.8 (Standard normal CF)**.** For $X \sim \mathcal{N}(0,1)$,

$$\varphi_X(t) \;=\; \int_{-\infty}^{\infty} e^{itx}\, \frac{1}{\sqrt{2\pi}}\, e^{-x^2/2}\, dx.$$

Completing the square in the exponent gives $-\frac{1}{2}(x - it)^2 - \frac{t^2}{2}$, so the integrand factors as a Gaussian kernel in $x - it$ times $e^{-t^2/2}$. The kernel integrates to one, leaving

$$\varphi_X(t) \;=\; e^{-t^2/2}, \qquad t \in \mathbb{R}. \tag{2}$$

This agrees with $M_X(it) = e^{(it)^2/2} = e^{-t^2/2}$, as expected.

Example 2.9 (Standard Cauchy CF)**.** For the standard Cauchy density $f(x) = 1/(\pi(1+x^2))$, a contour-integral computation (or Fourier-transform table; see Feller (1971, §XV.4)) gives

$$\varphi_X(t) \;=\; e^{-|t|}, \qquad t \in \mathbb{R}. \tag{3}$$

The Cauchy CF is well defined and continuous, even though its MGF is not.

Two algebraic properties make CFs computationally convenient. By independence, $\varphi_{X+Y}(t) = \varphi_X(t)\,\varphi_Y(t)$; by direct substitution, $\varphi_{aX+b}(t) = e^{itb}\,\varphi_X(at)$. Like the MGF, the CF determines the distribution uniquely, and an inversion formula recovers the CDF from φ_X (Billingsley, 1995, §26).

The deepest property of characteristic functions is the bridge they provide between pointwise convergence of φ_{X_n} and convergence in distribution of X_n.

Theorem 2.4 (Lévy continuity theorem)**.** Let X_n and X be random variables with characteristic functions φ_{X_n} and φ_X.

(1) If $X_n \xrightarrow{d} X$, then $\varphi_{X_n}(t) \to \varphi_X(t)$ for every $t \in \mathbb{R}$.

(2) Conversely, if $\varphi_{X_n}(t) \to \varphi(t)$ for every $t \in \mathbb{R}$ and φ is continuous at 0, then φ is the CF of some random variable X, and $X_n \xrightarrow{d} X$.

A full proof is in Billingsley (1995, Thm. 26.3). The theorem is the key to the central limit theorem proof in Section 2.5: it reduces a question about distributions to a calculus exercise on a single complex-valued function.

2.4 Modes of Convergence

A sequence of random variables can converge to a limit in several inequivalent ways. The three modes used in this report, from strongest to weakest, are the following.

Definition 2.10 (Three modes of convergence)**.** Let X_n, X be random variables on a common probability space.

- $X_n \xrightarrow{\text{a.s.}} X$ (*almost surely*): $\mathbb{P}\big(\lim_n X_n = X\big) = 1$.
- $X_n \xrightarrow{\mathbb{P}} X$ (*in probability*): for every $\varepsilon > 0$, $\mathbb{P}(|X_n - X| > \varepsilon) \to 0$.
- $X_n \xrightarrow{d} X$ (*in distribution*): $F_{X_n}(x) \to F_X(x)$ at every continuity point of F_X.

Proposition 2.5 (Implication chain)**.** $X_n \xrightarrow{\text{a.s.}} X \implies X_n \xrightarrow{\mathbb{P}} X \implies X_n \xrightarrow{d} X$. The converses are false in general.

Sketch. For a.s. $\Rightarrow$ probability: pointwise convergence on a set of probability one implies that for every $\varepsilon > 0$ the events $\{|X_n - X| > \varepsilon\}$ shrink to a null set. For probability $\Rightarrow$ distribution: the distributional version follows from a standard argument with continuity points; see Casella and Berger (2002, Thm. 5.5.12). □

A canonical counterexample showing that probability does not imply almost-sure convergence:

Example 2.11 ($\xrightarrow{\mathbb{P}}$ but not $\xrightarrow{\text{a.s.}}$)**.** Let $X_n \in \{0, 1\}$ with $\mathbb{P}(X_n = 1) = 1/n$. Then $\mathbb{P}(|X_n| > \varepsilon) = 1/n \to 0$, so $X_n \xrightarrow{\mathbb{P}} 0$. But $\sum_n \mathbb{P}(X_n = 1) = \sum_n 1/n = \infty$, so by the second Borel-Cantelli lemma (Appendix A) the event $\{X_n = 1\}$ occurs infinitely often almost surely, and hence $X_n \overset{\text{a.s.}}{\not\longrightarrow} 0$.

A second standard fact will be useful in later sections: continuous functions preserve convergence.

Proposition 2.6 (Continuous mapping theorem)**.** If $X_n \xrightarrow{\text{a.s.}} X$ (respectively $\xrightarrow{\mathbb{P}}$, $\xrightarrow{d}$) and $g : \mathbb{R} \to \mathbb{R}$ is continuous on a set of $\mathbb{P}_X$-probability one, then $g(X_n) \xrightarrow{\text{a.s.}} g(X)$ (respectively $\xrightarrow{\mathbb{P}}$, $\xrightarrow{d}$).

Slutsky's theorem (also useful later) is recorded in Appendix A.

2.5 Limit Theorems: LLN and CLT

Two universality results anchor classical probability: averages of iid samples converge to their mean (law of large numbers), and properly normalized averages have a universal Gaussian fluctuation (central limit theorem). We prove both, building on the tools of Sections 2.3–2.4.

We first need two classical inequalities.

Theorem 2.7 (Markov's inequality)**.** If $Y \geq 0$ and $a > 0$, then $\mathbb{P}(Y \geq a) \leq \mathbb{E}[Y]/a$.

Proof. On the event $\{Y \geq a\}$ we have $Y \geq a$, so $Y\mathbf{1}_{\{Y\geq a\}} \geq a\mathbf{1}_{\{Y\geq a\}}$. Taking expectations, $\mathbb{E}[Y] \geq \mathbb{E}[Y\mathbf{1}_{\{Y\geq a\}}] \geq a\,\mathbb{P}(Y \geq a)$, and the claim follows. $\square$

Theorem 2.8 (Chebyshev's inequality)**.** If $\mathbb{E}[X] = \mu$ and $\mathrm{Var}(X) = \sigma^2 < \infty$, then for any $\varepsilon > 0$,

$$\mathbb{P}(|X - \mu| \geq \varepsilon) \;\leq\; \frac{\sigma^2}{\varepsilon^2}.$$

Proof. Apply Markov's inequality to the nonnegative variable $Y = (X - \mu)^2$ with $a = \varepsilon^2$: $\mathbb{P}(|X - \mu| \geq \varepsilon) = \mathbb{P}((X - \mu)^2 \geq \varepsilon^2) \leq \mathbb{E}[(X - \mu)^2]/\varepsilon^2 = \sigma^2/\varepsilon^2$. $\square$

These suffice to prove the weak law.

Theorem 2.9 (Chebyshev's weak law of large numbers)**.** Let $X_1, X_2, \ldots$ be iid with $\mathbb{E}[X_1] = \mu$ and $\mathrm{Var}(X_1) = \sigma^2 < \infty$. Let $\bar{X}_n = n^{-1}\sum_{i=1}^n X_i$. Then $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$.

Proof. By linearity, $\mathbb{E}[\bar{X}_n] = \mu$; by independence, $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$. Chebyshev gives

$$\mathbb{P}(|\bar{X}_n - \mu| \geq \varepsilon) \;\leq\; \frac{\mathrm{Var}(\bar{X}_n)}{\varepsilon^2} \;=\; \frac{\sigma^2}{n\varepsilon^2} \;\longrightarrow\; 0.$$

$\square$

A stronger statement holds without the finite-variance assumption.

Theorem 2.10 (Kolmogorov's strong law of large numbers)**.** If $X_1, X_2, \ldots$ are iid with $\mathbb{E}[|X_1|] < \infty$ and $\mathbb{E}[X_1] = \mu$, then $\bar{X}_n \xrightarrow{\text{a.s.}} \mu$.

A proof is given in Billingsley (1995, Thm. 22.1). Figure 2 shows the convergence concretely.

The central limit theorem refines the law of large numbers by saying *at what rate* the convergence happens, and *what the fluctuations look like.*

Theorem 2.11 (Central limit theorem)**.** Let $X_1, X_2, \ldots$ be iid with $\mathbb{E}[X_1] = \mu$ and $0 < \mathrm{Var}(X_1) = \sigma^2 < \infty$. Then

$$Z_n \;:=\; \frac{\sqrt{n}\,(\bar{X}_n - \mu)}{\sigma} \;\xrightarrow{d}\; \mathcal{N}(0,1).$$

Proof. Let $Y_i = (X_i - \mu)/\sigma$. Then $\mathbb{E}[Y_i] = 0$, $\mathrm{Var}(Y_i) = 1$, and $Z_n = n^{-1/2}\sum_{i=1}^n Y_i$. Let $\varphi(t) = \mathbb{E}[e^{itY_1}]$. A second-order Taylor expansion of e^{iu} at $u = 0$ gives $e^{iu} = 1 + iu - u^2/2 + o(u^2)$; substituting $u = tY_1$ and taking expectations using $\mathbb{E}[Y_1] = 0$, $\mathbb{E}[Y_1^2] = 1$,

$$\varphi(t) \;=\; 1 \;-\; \tfrac{1}{2}t^2 \;+\; o(t^2) \qquad (t \to 0).$$

By independence, the CF of Z_n factors:

$$\varphi_{Z_n}(t) \;=\; \Big(\varphi\big(t/\sqrt{n}\big)\Big)^n \;=\; \Big(1 - \tfrac{t^2}{2n} + o(1/n)\Big)^n.$$

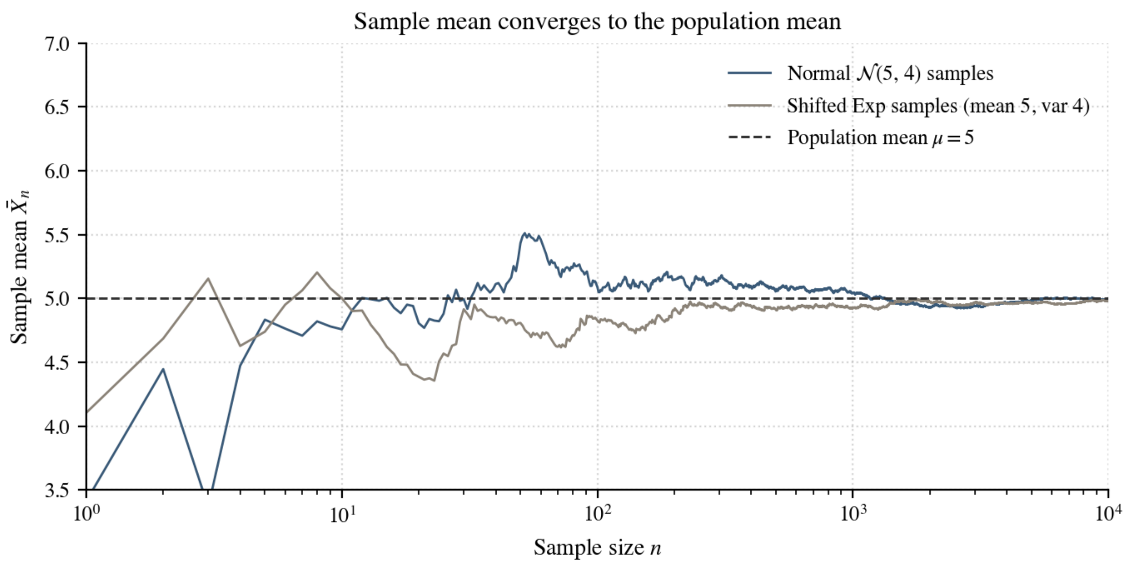

Figure 2: The sample mean $\bar{X}_n$ converges to the population mean $\mu = 5$ for two distributions with very different shapes but the same mean and variance: a normal $\mathcal{N}(5, 4)$ and an exponential rescaled and shifted to also have mean 5 and variance 4. The exponential samples fluctuate more in the early stages because their distribution is heavily skewed, but both averages settle near 5 by $n = 10000$. Generated by `notebooks/01_lln_clt_demos.py`.

The right side converges to $e^{-t^2/2}$ as $n \to \infty$ by the standard limit $(1 + a_n/n)^n \to e^a$ when $a_n \to a$. The function $e^{-t^2/2}$ is the CF of $\mathcal{N}(0, 1)$ by Example 2.8, so the Lévy continuity theorem (Theorem 2.4) yields $Z_n \xrightarrow{d} \mathcal{N}(0, 1)$. □

Figure 3 illustrates the CLT for exponential samples: even though the underlying distribution is sharply skewed, the histogram of the standardized sample mean Z_n approaches the standard normal density as n grows.

2.6 Random Vectors and the Multivariate Normal Distribution

The objects of this report are sample covariance matrices, which act on random vectors rather than random scalars. We collect the multivariate analogues of the previous subsections here.

Definition 2.12 (Random vector)**.** A *random vector* is a function $\boldsymbol{X} = (X_1, \ldots, X_p)^\top : \Omega \to \mathbb{R}^p$ whose components are random variables on a common probability space. Its *joint CDF* is $F_{\boldsymbol{X}}(\boldsymbol{x}) = \mathbb{P}(X_1 \le x_1, \ldots, X_p \le x_p)$.

The mean and covariance of $\boldsymbol{X}$ collect the first two moments of all coordinates.

Definition 2.13 (Mean vector and covariance matrix)**.** The *mean vector* is $\boldsymbol{\mu} = \mathbb{E}[\boldsymbol{X}] = (\mathbb{E}[X_1], \ldots, \mathbb{E}[X_p])^\top$. The *covariance matrix* is

$$\Sigma \;=\; \mathrm{Cov}(\boldsymbol{X}) \;=\; \mathbb{E}\big[(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^\top\big],$$

a $p \times p$ matrix with entries $\Sigma_{ij} = \mathrm{Cov}(X_i, X_j)$.

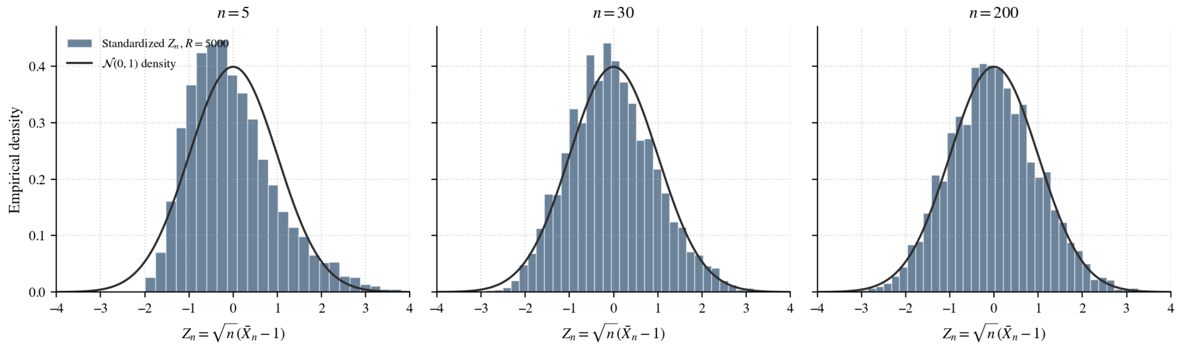

Figure 3: Central limit theorem for iid Exponential(1) samples. Each panel shows a histogram of $R = 5000$ standardized sample means $Z_n = \sqrt{n}(\bar{X}_n - 1)$ at increasing n. The histogram is sharply skewed at $n = 5$, visibly less so at $n = 30$, and indistinguishable from the standard normal density (solid line) by $n = 200$. Generated by `notebooks/01_lln_clt_demos.py`.

Two structural facts:

Proposition 2.12 (Σ is symmetric and PSD)**.** For any random vector with finite second moments, the covariance matrix Σ is symmetric and positive semidefinite. It is positive definite if and only if no nontrivial linear combination $\boldsymbol{v}^\top \boldsymbol{X}$ ($\boldsymbol{v} \neq 0$) is almost surely constant.

Proof. Symmetry is immediate from $\Sigma_{ij} = \mathrm{Cov}(X_i, X_j) = \mathrm{Cov}(X_j, X_i) = \Sigma_{ji}$. For PSD, fix any $\boldsymbol{v} \in \mathbb{R}^p$ and let $Y = \boldsymbol{v}^\top \boldsymbol{X}$. Then $Y - \mathbb{E}[Y] = \boldsymbol{v}^\top(\boldsymbol{X} - \boldsymbol{\mu})$, so

$$\mathrm{Var}(Y) \;=\; \mathbb{E}\big[(\boldsymbol{v}^\top(\boldsymbol{X} - \boldsymbol{\mu}))^2\big] \;=\; \boldsymbol{v}^\top\, \mathbb{E}\big[(\boldsymbol{X} - \boldsymbol{\mu})(\boldsymbol{X} - \boldsymbol{\mu})^\top\big]\, \boldsymbol{v} \;=\; \boldsymbol{v}^\top \Sigma \boldsymbol{v}.$$

Since $\mathrm{Var}(Y) \geq 0$, we have $\boldsymbol{v}^\top \Sigma \boldsymbol{v} \geq 0$ for every $\boldsymbol{v}$, which is PSD. Equality holds for some $\boldsymbol{v} \neq 0$ precisely when $\boldsymbol{v}^\top \boldsymbol{X}$ is almost surely constant, which is the failure of positive definiteness. □

The PSD identity $\boldsymbol{v}^\top \Sigma \boldsymbol{v} = \mathrm{Var}(\boldsymbol{v}^\top \boldsymbol{X})$ is the structural fact that organizes the analysis of sample covariance matrices: every quadratic form on a covariance matrix is a variance, and every spectral statement about Σ translates to a statement about variances of linear combinations of $\boldsymbol{X}$.

Definition 2.14 (Multivariate normal)**.** A random vector $\boldsymbol{X} \in \mathbb{R}^p$ is *multivariate normal* with mean $\boldsymbol{\mu}$ and covariance Σ, written $\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, if its characteristic function is

$$\varphi_{\boldsymbol{X}}(\boldsymbol{t}) \;=\; \exp\Big(i\, \boldsymbol{t}^\top \boldsymbol{\mu} \;-\; \tfrac{1}{2}\, \boldsymbol{t}^\top \Sigma \boldsymbol{t}\Big), \qquad \boldsymbol{t} \in \mathbb{R}^p. \tag{4}$$

When Σ is positive definite, $\boldsymbol{X}$ admits the density

$$f_{\boldsymbol{X}}(\boldsymbol{x}) \;=\; \frac{1}{(2\pi)^{p/2}\, |\Sigma|^{1/2}}\, \exp\Big(-\tfrac{1}{2}\, (\boldsymbol{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\boldsymbol{x} - \boldsymbol{\mu})\Big).$$

Equation (4) is taken as the definition because it remains valid when Σ is singular (a case that arises when components are linearly dependent), whereas the density form requires Σ to be invertible.

The multivariate normal is closed under affine transformations. If $\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$ and $\boldsymbol{A} \in \mathbb{R}^{q\times p}$, $\boldsymbol{b} \in \mathbb{R}^q$, then $\boldsymbol{AX} + \boldsymbol{b} \sim \mathcal{N}(\boldsymbol{A\mu} + \boldsymbol{b},\, \boldsymbol{A}\Sigma\boldsymbol{A}^\top)$. Verification is a one-line CF computation using (4).

A second special property of Gaussians is that uncorrelated components are independent. For general distributions, $\mathrm{Cov}(X_1, X_2) = 0$ does not imply independence; for the multivariate normal it does, because the density factors as soon as Σ is diagonal. Figure 4 shows the density contours for three correlation values.

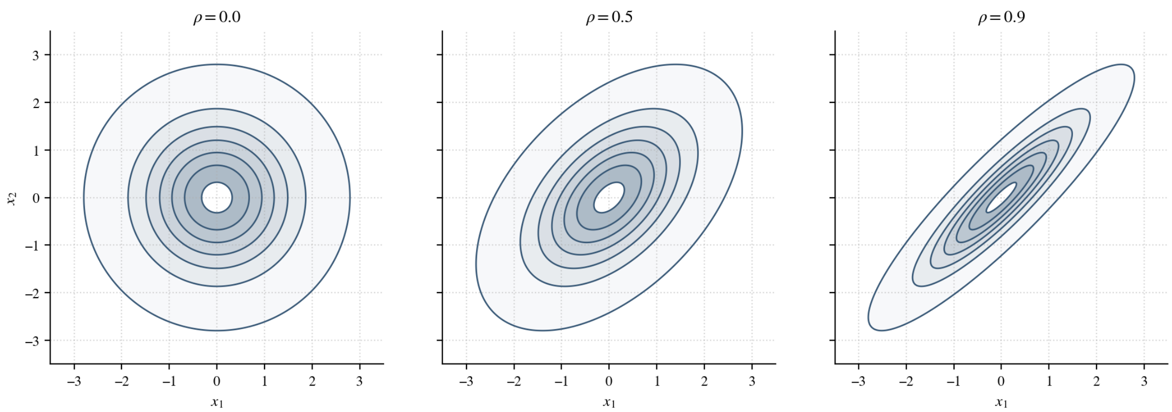

Figure 4: Density contours of the centered bivariate normal with unit variances and correlation $\rho \in \{0,\, 0.5,\, 0.9\}$. At $\rho = 0$ the contours are circles and the components are independent; as ρ increases, the contours elongate along the line $x_2 = x_1$, reflecting positive linear association. Generated by `notebooks/01_lln_clt_demos.py`.

The multivariate analogue of the central limit theorem follows from the univariate version by the *Cramér-Wold device*: a sequence of random vectors converges in distribution if and only if every linear combination converges in distribution. Applied to iid centered random vectors with covariance Σ,

$$\sqrt{n}\,(\bar{\boldsymbol{X}}_n - \boldsymbol{\mu}) \xrightarrow{d} \mathcal{N}(\boldsymbol{0},\, \Sigma).$$

This is the fundamental result that powers every fixed-p sample covariance computation in Section 3, and it sets up the contrast with the high-dimensional regime, where $\sqrt{n}$-Gaussian fluctuations no longer suffice to describe the spectral behavior of $\widehat{\Sigma}_n$.

3 Sample Covariance Matrices

[TODO] *Opening paragraph: the sample covariance matrix $\widehat{\Sigma}_n$ is the central object of multivariate statistics. In low dimensions (p fixed, $n \to \infty$) it converges almost surely to the population covariance Σ and its eigenvalues converge by continuity. In high dimensions ($p, n \to \infty$ with $p/n \to y > 0$) this convergence fails dramatically: the empirical spectrum spreads out into a non-trivial bulk density. This section develops both regimes, building from the foundations laid in Section 2.*

3.1 Population and Sample Covariance, Spectral Theorem, Positive Semidefiniteness

[TODO] *Content to fill in (from v5 + Joseph Section 3 intro): - Definition: sample covariance matrix $\widehat{\Sigma}_n = \frac{1}{n}\sum_{i=1}^n \boldsymbol{X}_i\boldsymbol{X}_i^\top$ (mean-zero case) and the unbiased $\frac{1}{n-1}$ variant. - Matrix form: $\widehat{\Sigma}_n = \frac{1}{n}\boldsymbol{X}\boldsymbol{X}^\top$ for $\boldsymbol{X} \in \mathbb{R}^{p\times n}$. - Spectral theorem for symmetric matrices: $A = Q\Lambda Q^\top$. - PSD characterization: A is PSD iff all eigenvalues are nonneg (proof). - Use the diagonalization to compute quadratic forms; foreshadow why eigenvalues of $\widehat{\Sigma}_n$ are the meaningful quantities.*

3.2 Low Dimensions: Wishart, Eigenvalue Repulsion, and Fluctuations

[TODO] *Content to fill in (from v5 + Joseph Section 3.1):*

(a) Almost sure convergence of the sample covariance. - Theorem: if $\boldsymbol{X}_i$ iid with $\mathbb{E}\boldsymbol{X}_1 = 0$ and $\mathrm{Cov}(\boldsymbol{X}_1) = \Sigma$, then $\widehat{\Sigma}_n \xrightarrow{a.s.} \Sigma$ entrywise by SLLN. - Corollary: $\lambda_i(\widehat{\Sigma}_n) \xrightarrow{a.s.} \lambda_i(\Sigma)$ by continuity of eigenvalues (proof sketch via continuity of the characteristic polynomial and continuity of its roots). Cite Genzer (2025, Prop. 3.1, 3.3).

(b) Wishart distribution. - Definition: $\boldsymbol{S} = \sum_{i=1}^n \boldsymbol{X}_i\boldsymbol{X}_i^\top \sim W_p(n, \Sigma)$ for iid Gaussian $\boldsymbol{X}_i \sim \mathcal{N}(0, \Sigma)$. - Wishart density formula. - Theorem: for $p = 1$, $W_1(n, 1)$ collapses to χ^2_n (proof by plugging in).

(c) 2D Wishart and eigenvalue repulsion. - Joint density of ordered eigenvalues of $W_2(n, I_2)$ contains the factor $\lambda_2 - \lambda_1$, producing eigenvalue repulsion. - Simulation: scatter of (λ_1, λ_2) and histogram of $\lambda_2 - \lambda_1$ (notebook 02). Include figure: 2D Wishart eigenvalue scatter.

(d) Joint CLT for entries of $\widehat{\Sigma}_n$ when $\Sigma = I_2$. - Statement and proof via multivariate CLT (variances 2, 2, 1 for diagonal and off-diagonal entries). - Consequence: matrix-valued CLT $\sqrt{n}(\widehat{\Sigma}_n - I_2) \xrightarrow{d} G$ where G has GOE structure.

(e) Top eigenvalue fluctuation: repeated vs. distinct cases. - Repeated ($\Sigma = I_2$): $\sqrt{n}(\lambda_2(\widehat{\Sigma}_n) - 1) \xrightarrow{d} \lambda_2(G)$, not *Gaussian. - Distinct ($\Sigma = \mathrm{diag}(1, 2)$): delta-method gives Gaussian fluctuations with variances 2 and 8. - Simulation: Q-Q plots for both cases (notebook 03). Include figures: Q-Q repeated, Q-Q distinct.*

(f) Eigenvector convergence. - Distinct case: $\widehat{\boldsymbol{u}}_\ell \xrightarrow{a.s.} \boldsymbol{e}_\ell$ with sign convention. - Repeated case: eigenvectors fail to converge; explicit counterexample with rotating O_n. This previews why spectral separation is essential for eigenvector recovery, a theme that returns in Section 4.

3.3 High Dimensions: Marchenko-Pastur and the Tracy-Widom Edge

[TODO] *Content to fill in (from v5 high-dim section + Joseph Section 3.2):*

(a) The high-dimensional regime. - Setup: $p, n \to \infty$ with $p/n \to y \in (0, \infty)$. - Empirical spectral distribution $F_{\widehat{\Sigma}_n}(\lambda) = \frac{1}{p}\#\{i : \lambda_i(\widehat{\Sigma}_n) \leq \lambda\}$.

(b) Marchenko-Pastur law. - Theorem (Marchenko-Pastur 1967): for null $\Sigma = I_p$ and $0 < y < 1$, $F_{\widehat{\Sigma}_n} \xrightarrow{d} \mathrm{MP}_y$ with support $[(1-\sqrt{y})^2, (1+\sqrt{y})^2]$ and density $f_{\mathrm{MP}}(\lambda) = \frac{1}{2\pi y\lambda}\sqrt{(\lambda_+ - \lambda)(\lambda - \lambda_-)}$. - For $y \geq 1$, an atom at zero of mass $1 - 1/y$. - Convergence of $\lambda_{\max}(\widehat{\Sigma}_n) \xrightarrow{a.s.} (1+\sqrt{y})^2$. - Simulation: histogram against MP density at $y = 1/2$ (notebook 04). Include figure: MP density vs. simulation.

(c) Tracy-Widom edge heuristic. - Goal: explain why edge fluctuations live on scale $p^{-2/3}$, not $n^{-1/2}$. - Heuristic: square-root density near the edge, $f_{\mathrm{MP}}(\lambda) \sim C(\lambda_+ - \lambda)^{1/2}$ as $\lambda \uparrow \lambda_+$, combined with the requirement that the mass to the right of $\lambda_{\max}$ be of order $1/p$, yields $\lambda_+ - \lambda_{\max} \sim p^{-2/3}$. - Statement (informal): $p^{2/3}(\lambda_{\max} - \lambda_+) \xrightarrow{d} \mathrm{TW}_1$ after model-dependent constants. - Remark on large-deviation heuristic confirming the same scaling from left and right tails. Cite Tracy and Widom (1996); Johnstone (2001).

4 Spiked Covariance Matrices and the BBP Transition

[TODO] *Opening paragraph: the Marchenko-Pastur law of Section 3.3 described the null case $\Sigma = I_p$, where all population eigenvalues are equal. In applications the population covariance carries structure: a small number of distinguished directions of high variance riding above a flat noise baseline. The* spiked covariance model *of Johnstone (2001) formalizes this,*

$$\Sigma = \operatorname{diag}(\alpha_1, \ldots, \alpha_m, 1, \ldots, 1),$$

with m fixed as $p \to \infty$. The central question is when these population spikes are detectable from the sample spectrum, and when the sample eigenvectors carry meaningful information about the population spike directions. The answer is a clean phase transition.

4.1 Johnstone's Spiked Model and the Spike-Location Map

[TODO] *Content to fill in (from v5 + Yao-Zheng-Bai Ch. 11 + Gustavo notes): - Definition: Johnstone's spiked model $\Sigma = \operatorname{diag}(\alpha_1, \ldots, \alpha_m, 1, \ldots, 1)$ with $H = \delta_1$ for the bulk limiting distribution. - Definition: spike-location map $\Psi(\alpha) = \alpha + \frac{y\alpha}{\alpha - 1}$. - Compute $\Psi'(\alpha) = 1 - \frac{y}{(\alpha-1)^2}$. - Lemma: $\Psi'(\alpha) > 0 \iff |\alpha - 1| > \sqrt{y}$; for large spikes $\alpha > 1$ this becomes $\alpha > 1 + \sqrt{y}$. - Verify boundary behavior: $\Psi(1 \pm \sqrt{y}) = (1 \pm \sqrt{y})^2$, so the spike map agrees with the MP edges at the threshold.*

4.2 The BBP Phase Transition for Eigenvalues

[TODO] *Content to fill in: - Theorem (Yao-Zheng-Bai Corollary 11.4 specialized to Johnstone): For a large spike $\alpha_j > 1 + \sqrt{y}$,*

$$\lambda_i(\widehat{\Sigma}_n) \xrightarrow{a.s.} \Psi(\alpha_j), \quad i \in J_j.$$

For $1 < \alpha_j \leq 1 + \sqrt{y}$,

$$\lambda_i(\widehat{\Sigma}_n) \xrightarrow{a.s.} (1 + \sqrt{y})^2.$$

Brief sketch (cite Baik and Silverstein (2006) for the proof). - Worked example: $y = 1/2$, so $1 + \sqrt{y} \approx 1.707$ and edge $(1 + \sqrt{y})^2 \approx 2.914$. For $\alpha = 2.5$, $\Psi(2.5) \approx 3.333$. For $\alpha = 1.4$, $\alpha < 1.707$ so the spike is absorbed into the edge. - Simulations (existing figures from v5): Figure: smoothed m=1 simulation, strong vs. weak spike (`smoothed_m1_strong_vs_weak.png`). Figure: raw top eigenvalue distributions (`m1_strong_spike.png`, `m1_weak_spike.png`). Figure (optional): $m = 2$ cases (`m2_two_strong_spikes.png`, `m2_mixed_spikes.png`). Cite Baik et al. (2005); Yao et al. (2015).

4.3 Fluctuations of Spiked Eigenvalues

[TODO] *Content to fill in (from Gustavo's promised sketch + v5): - Statement (fundamental spike, qualitative): for $\alpha > 1 + \sqrt{y}$ in the Gaussian Johnstone model,*

$$\sqrt{n}(\lambda_{\max}(\widehat{\Sigma}_n) - \Psi(\alpha)) \xrightarrow{d} \mathcal{N}(0, \sigma^2_{\alpha,y}).$$

Cite Yao et al. (2015, Thm. 11.11) (fundamental spikes) and Féral and Péché (2009, Thm. 1.6) (cleanest diagonal case). - Heuristic: a separated eigenvalue is a smooth functional of $\widehat{\Sigma}_n$, so a delta-method argument gives $\sqrt{n}$ Gaussian fluctuations (compare with Section 3.2(e)). - Non-fundamental spike: top eigenvalue attaches to the edge, $\lambda_{\max} - \lambda_+ = O_p(p^{-2/3})$, and the fluctuation is expected to be Tracy-Widom type (cite Johnstone (2001)). - Honest note: the critical case $\alpha = 1 + \sqrt{y}$ is not resolved in our current treatment; cite open literature. - Simulation: histogram and Gaussian Q-Q for strong-spike fluctuations (notebook 06).

4.4 Eigenvector Alignment Beyond the Threshold

[TODO] *Content to fill in (from v5 + Yao-Zheng-Bai Thm 11.5/Cor 11.6): - Set-up: in the one-spike model, the population spike direction is $\boldsymbol{e}_1$. Let $\widehat{\boldsymbol{u}}_1$ be the top sample eigenvector. The squared overlap $|\langle \widehat{\boldsymbol{u}}_1, \boldsymbol{e}_1 \rangle|^2 = \cos^2\theta$ removes the sign ambiguity. - Qualitative theorem (Yao-Zheng-Bai Thm. 11.5, Cor. 11.6): strong spike $\Rightarrow$ overlap converges to a positive constant; weak spike $\Rightarrow$ overlap converges to zero. - The one-spike Johnstone overlap formula:*

$$\rho(\alpha, y) = \frac{1 - \dfrac{y}{(\alpha-1)^2}}{1 + \dfrac{y}{\alpha - 1}}.$$

*Positive precisely when $\alpha > 1+\sqrt{y}$. - Simulation (figure already in v5/figures/): **`eigenvector_overlap_dis`** shows strong-spike overlaps clustered around 0.55–0.65 and weak-spike overlaps near 0. Include and discuss.*

4.5 Sketch of the Connection to Spectral Clustering

[TODO] *Content to fill in (from Gustavo's email + v5 + von Luxburg 2007): - Setup: data points $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_n \in \mathbb{R}^p$, kernel similarity $K_{ij} = f(\|\boldsymbol{x}_i - \boldsymbol{x}_j\|^2/p)$. - Definition: degree matrix $D = \operatorname{diag}(K\mathbf{1})$ and normalized Laplacian-style kernel $L = nD^{-1/2}KD^{-1/2}$. - Heuristic decomposition: L = noise bulk + low-rank class signal, in direct analogy with $\widehat{\Sigma}_n$ = noise + spike. - Conclusion: class structure is recoverable from L precisely when the signal-to-noise ratio crosses an analogue of the BBP threshold; the informative eigenvectors of L then provide the clustering. Cite von Luxburg (2007) for the algorithmic side and Couillet and Benaych-Georges (2016) for the high-dimensional analysis. - Toy simulation: notebook 08, a two-class Gaussian mixture and the top non-trivial eigenvector of L.*

5 Conclusion and Outlook

[TODO] *Content to fill in: (a) Summary table: the strong vs. weak spike contrast, three rows: eigenvalue limit, eigenvector overlap, fluctuation scale.*

— Spike strength — Eigenvalue limit — Overlap with $\boldsymbol{e}_1$ *— Fluctuation — ————— —-——————————-———————————————-————————-— — Strong* $\alpha > 1+\sqrt{y}$ *—* $\Psi(\alpha)$ *—* $\rho(\alpha, y) > 0$ *— Gaussian,* $\sqrt{n}$ *— — Weak* $1 < \alpha \leq 1+\sqrt{y}$ *—* $(1+\sqrt{y})^2$ *— 0 — Edge / Tracy-Widom,* $p^{-2/3}$ *—*

(b) One-paragraph synthesis: the BBP phase transition is the cleanest quantitative example of when high-dimensional inference recovers structure and when it does not. It explains, in a controlled toy model, why methods like PCA and spectral clustering succeed only when signal exceeds a threshold determined by the aspect ratio.

(c) Outlook: open questions to flag. - The critical case $\alpha = 1+\sqrt{y}$*. The current literature does not yet contain a clean limit theorem; cite open work. - Exact variance* $\sigma^2_{\alpha,y}$ *in the strong-spike CLT. - Beyond Johnstone's case: general spiked models with non-flat bulk (cite Baik and Silverstein (2006); Benaych-Georges and Nadakuditi (2011)). - Rigorous spectral clustering derivations beyond the toy model; cite Couillet and Benaych-Georges (2016).*

(d) Closing sentence: the goal of this thesis was to develop the foundational picture from one end of the dimension spectrum to the other; the open questions above point to where the field is going.

A Auxiliary Results

[TODO] *Content to fill in:*

A.1 Borel-Cantelli Lemmas. - First Borel-Cantelli: if $\sum_n \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 0$. - Second Borel-Cantelli (independence version): if events are independent and $\sum_n \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(A_n \text{ i.o.}) = 1$. - Used in Section 2.4 for the convergence-in-probability but not-almost-surely counterexample.

A.2 Slutsky's theorem. - Statement and a one-paragraph proof or reference to Casella and Berger (2002, Thm. 5.5.17).

A.3 Continuous mapping theorem. - Statement: if $X_n \xrightarrow{d} X$ and g is continuous on a set of $\mathbb{P}_X$-probability one, then $g(X_n) \xrightarrow{d} g(X)$.

A.4 Multivariate delta method. - Statement and reference; used in Sections 3.2 and 4.3.

A.5 Characteristic functions and the Lévy continuity theorem. - Definition: $\varphi_X(t) = \mathbb{E}[e^{itX}]$; always exists. - Examples: $\mathcal{N}(0,1)$ gives $e^{-t^2/2}$, Cauchy gives $e^{-|t|}$. - Lévy: $X_n \xrightarrow{d} X \iff \varphi_{X_n}(t) \to \varphi_X(t)$ pointwise, with continuity at 0 sufficient for the converse. - Proof of the classical CLT via $(1 + t/n)^n \to e^t$ expansion.

B Simulation Methodology

[TODO] *Content to fill in:*

B.1 General reproducibility conventions. - All notebooks use Python 3, with `numpy`, `matplotlib`, and `scipy.stats`. - Random number generation: `rng = np.random.default_rng(seed)`, with explicit `seed` at the top of every notebook. - Eigendecomposition: `np.linalg.eigh` on symmetric inputs; the top eigenpair is taken as `evals[-1]`, `evecs[:, -1]`. - Plot styling: shared helper module `notebooks/thesis_style.py` applies the six-color palette and sans-serif math labels.

B.2 Per-notebook parameter table. - Notebook 01 (LLN/CLT demos): $n \in \{100, 1000, 10000\}$*, 500 trials. - Notebook 02 (Wishart and repulsion):* $n = 100$*,* $p = 2$*, 5000 trials. - Notebook 03 (fixed-dim fluctuations):* $n = 500$*, 5000 trials, both* $\Sigma = I_2$ *and* $\Sigma = \mathrm{diag}(1, 2)$*. - Notebook 04 (Marchenko-Pastur & TW):* $n = 1000$*,* $y = 1/2$*, 1 trial for bulk density, 1000 trials for top eigenvalue. - Notebook 05 (spiked eigenvalues, BBP):* $y = 1/2$*,* $n \in \{100, 200, 400, 800, 1200\}$*, 200 trials per* n*,* $\alpha \in \{1.4, 2.5\}$*. - Notebook 06 (fluctuation Q-Q):* $n = 1200$*,* $y = 1/2$*, 500 trials,* $\alpha = 2.5$*. - Notebook 07 (eigenvector overlap):* $n = 1200$*,* $y = 1/2$*, 500 trials,* $\alpha \in \{1.4, 2.5\}$*. - Notebook 08 (spectral clustering toy): two-class Gaussian mixture,* $n = 600$ *per class,* $p \in \{50, 200\}$*.*

B.3 Numerical caveats. - For the strong-spike fluctuation Q-Q plot, we standardize empirically (subtract sample mean, divide by sample standard deviation) because the exact variance $\sigma^2_{\alpha,y}$ *requires constants from Yao et al. (2015, Thm. 11.11). - For the weak spike, we do not attempt an exact Tracy-Widom fit because the finite-sample centering and scaling constants are delicate; we report the qualitative shape of the histogram and defer the rigorous fit to future work. - The realized aspect ratio* p/n *differs from the target* y *when* yn *is not integer; figures use the realized ratio in the bulk-edge formula* $(1 + \sqrt{p/n})^2$ *for consistency.*

C Selected Code Listings

[TODO] *Include 4 to 6 of the most informative snippets from the notebooks (not full notebooks). Each listing should be self-contained at the function level, captioned, and referenced from the main text where the corresponding figure appears.*

Suggested listings to include here:

- *C.1 One-spike Johnstone simulation (driving function for Section 4.2). Source:* `notebooks/05_spiked`
- *C.2 Eigenvector overlap simulation (Section 4.4). Source:* `notebooks/07_eigenvector_overlap.ipy`
- *C.3 Strong-spike fluctuation simulation (Section 4.3). Source:* `notebooks/06_fluctuation_qq_plots`
- *C.4 Marchenko-Pastur bulk simulation (Section 3.3). Source:* `notebooks/04_marchenko_pastur_trac`

Each listing will be wrapped in a `lstlisting` *environment using the* `thesisPython` *style defined in* `style/thesis.sty`*, with a caption and a* `label` *for cross-referencing from the main text.*

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